FLasher: Seasonal examples

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Examples

- Demersal stock with recruitment at age 1 spawning in 1^{st} of 4 seasons
- Pelagic stock with recruitment at age 0 that spawns in 2^{nd} of 4 seasons
- Temperate tuna with two sexes spawning in 1^{st} of 4 seasons
- Tropical tuna that spawns each of 4 seasons
- Flatfish stock with sexual dimorphism

Load requitred 'FLR' libraries

```
library(FLCore)
library(FLBRP)
library(ggplotFL)
```

```
library(devtools)
install_github("flr/FLasher",ref="fix_seasons")
```

```
library(FLasher)
```

Demersal stock

Example is based on the North Sea Plaice 'ple4' object with recruitment at age 1 that spawns in 1^{st} of 4 season.

Load the object,

```
data(ple4)
```

fit a stock-recruitment relationship,

```
ple4.sr=fmle(as.FLSR(ple4,model="bevholt"),control=list(silent=TRUE))
```

and estimate reference points using 'FLBRP'

```
ple4.eq=FLBRP(ple4,sr=ple4.sr)

plot(ple4.eq, obs=TRUE)
```

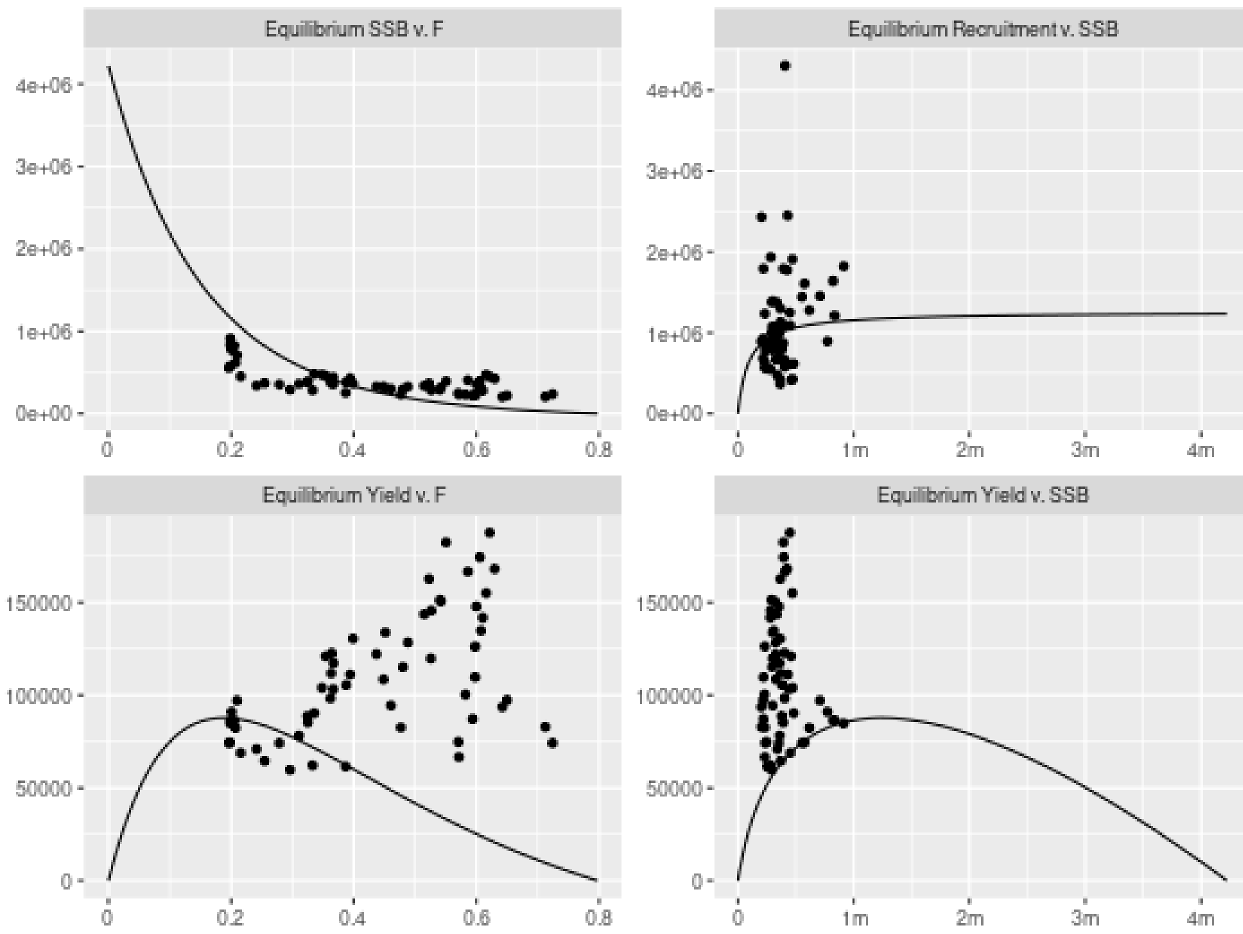


Figure 1 Equilbriium curves with MSY reference points.

Historically the stock has been overexploited, although recruitment has not been impaired, since steepness is high. Recently F has been reduced to F_{MSY} and the stock is recovering to B_{MSY}, although catches are stable.

```
plot(ple4, metrics=list(Recruits=rec, SSB=ssb, Catch=catch, F=fbar)) +
   geom_flpar(data=FLPars(Recruits=FLPar(RMSY=refpts(ple4.eq)["msy","rec"]),
                          SSB     =FLPar(BMSY=refpts(ple4.eq)["msy","ssb"]),
                          Catch   =FLPar( MSY=refpts(ple4.eq)["msy","yield"]),
                          F       =FLPar(FMSY=refpts(ple4.eq)["msy","harvest"])), x=1960)+
  theme_bw()+
  xlab("Year")
```

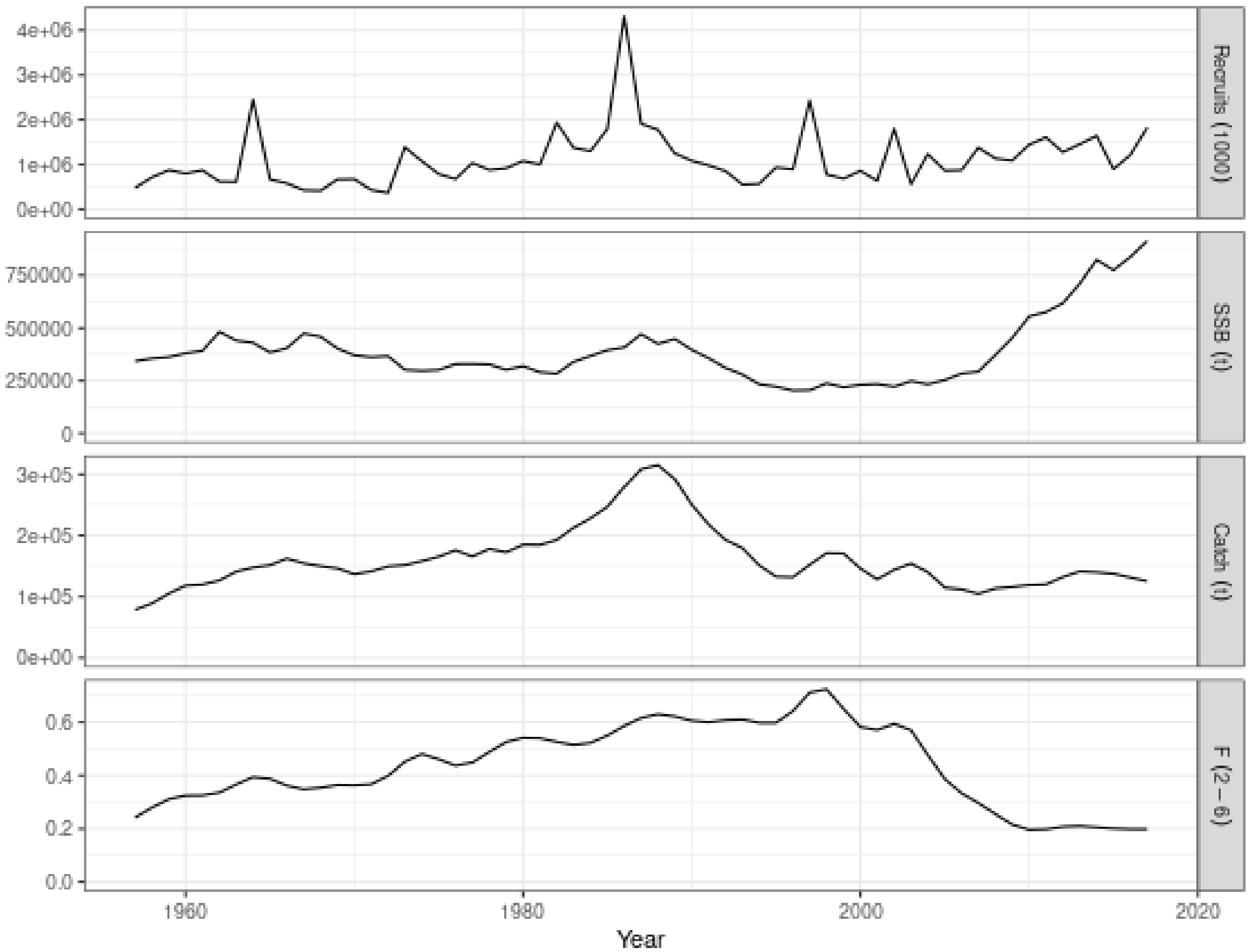


Figure 2 Time series relative to MSY benchmarks

Seasons

To create a seasonal 'FLStock' expand the 4^{th} dimension, by using 'expand'

```
ple4.4=expand(ple4,season=1:4)
```

Divide natural mortality and fishing mortality equally across seasons

```
m(      ple4.4)=m(ple4.4)/dim(ple4.4)[4]
harvest(ple4.4)=harvest(ple4.4)/dim(ple4.4)[4]
```

To ensure that the initial stock vector is consistent with the new structure, update the stock numbers values in the 1^{st} year

```
stock.n(ple4.4)[,1,,2]=stock.n(ple4.4)[,1,,1]*exp(-m(ple4.4)[,1,,1]-harvest(ple4.4)[,1,,1])
stock.n(ple4.4)[,1,,3]=stock.n(ple4.4)[,1,,2]*exp(-m(ple4.4)[,1,,2]-harvest(ple4.4)[,1,,2])
stock.n(ple4.4)[,1,,4]=stock.n(ple4.4)[,1,,3]*exp(-m(ple4.4)[,1,,3]-harvest(ple4.4)[,1,,3])
```

Perform a projection from the 1^{st} year onwards to ensure consistency of stock numbers with the seasonal model F and M, specify initial recruitment based on estimates in the original 'ple4' object.

```
sr=predictModel(model=geomean()$model,
                params=FLPar(rec(ple4)[drop=T],dimnames=list(params="a",year=dimnames(rec(ple4))$year,s
params( sr)[,,-1]=NA
```

and set the target F in each season and year.

```
control=as(FLQuants(fbar=fbar(ple4.4)[,-1]), 'fwdControl')

ple4.4=fwd(ple4.4, control=control, sr=sr)[,-1]
```

Compare the annual and seasonal models. SSB is the same, while recruitment in the 1_{st} season matches in the spawning season in both models. A decline is seen in the seasons after spawning. The seasonal values of catch and F shown in the plot are a 4_{th} of the annual values.

```
mat(ple4.4)[,,,-1]=NA
plot(FLStocks("Annual"=ple4,"Seasonal"=window(ple4.4,end=2017)))+
  scale_colour_manual("Model",values=c("blue","purple"))+
  theme_bw()+
  theme(legend.position="bottom")
```

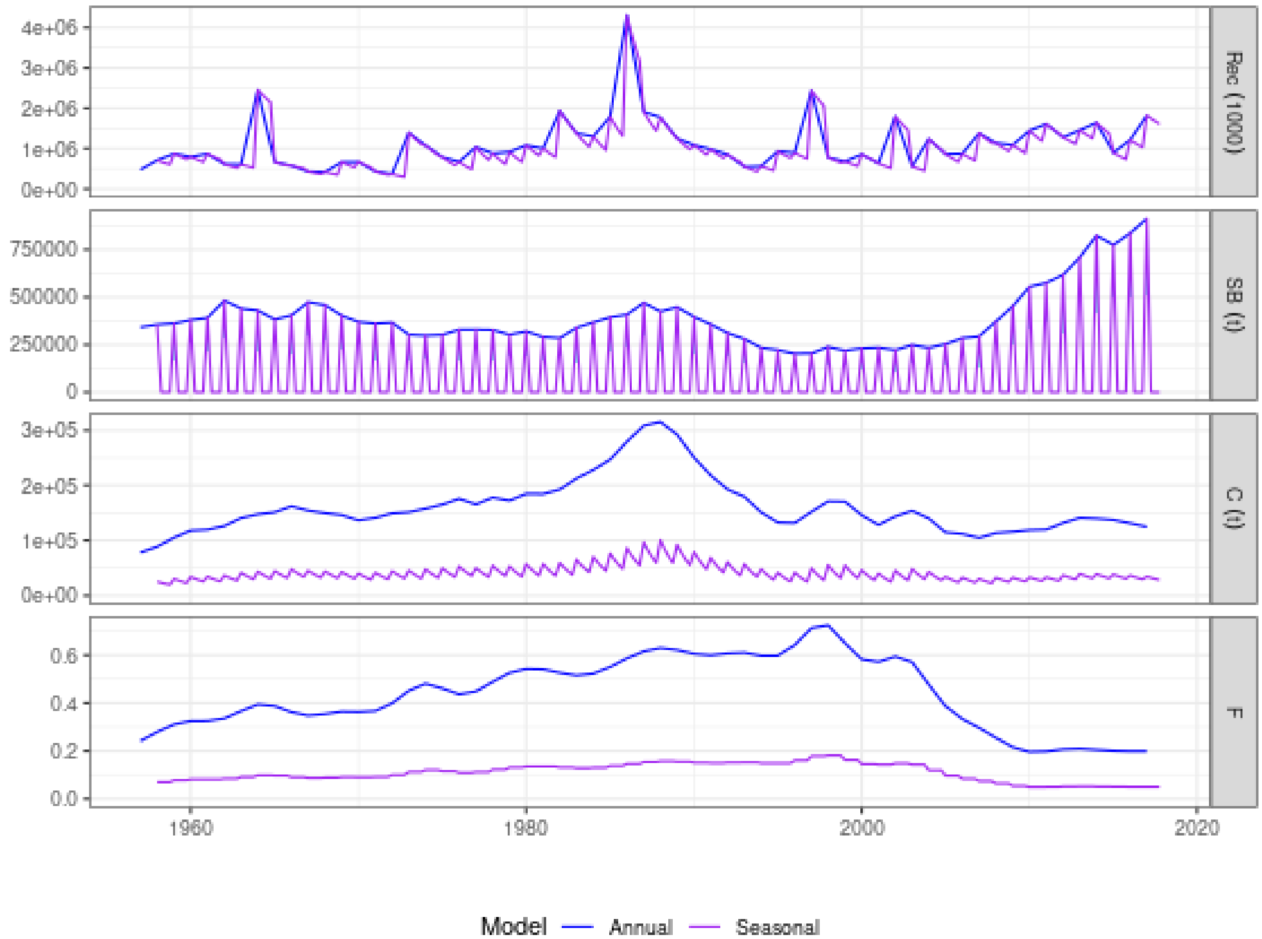


Figure 3 Comparison of amnnual and seasonal models.

That they match can be demonstrated by summing them over seasons

```
ggplot(as.data.frame(FLQuants("ple4"=catch(ple4),"Seasonal"=apply(catch(ple4.4),2,sum))))+
  geom_line(aes(year,data,col=qname))+
  theme_bw()+
  xlab("Year")+ylab("Fbar")
```

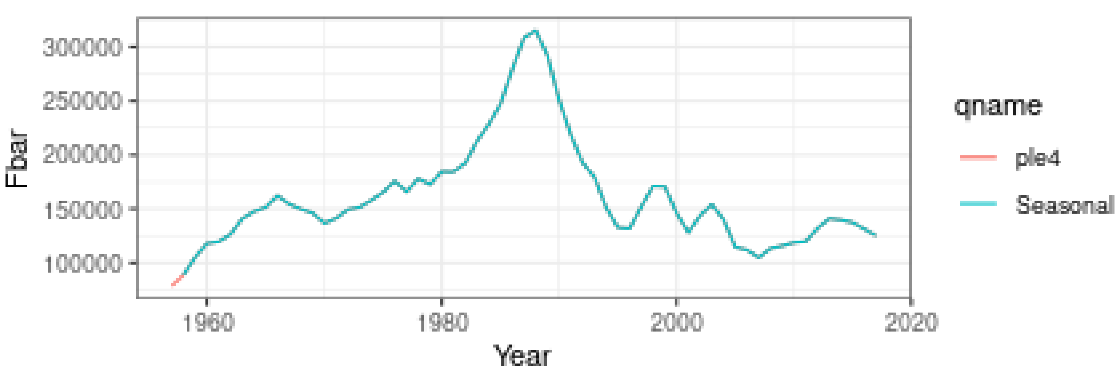


Figure 4 Comparison of catches

```
ggplot(as.data.frame(FLQuants("ple4"=fbar(ple4),"Seasonal"=apply(fbar(ple4.4),2,sum))))+
  geom_line(aes(year,data,col=qname))+
  theme_bw()+
```

```
  xlab("Year")+ylab("SSB")
```

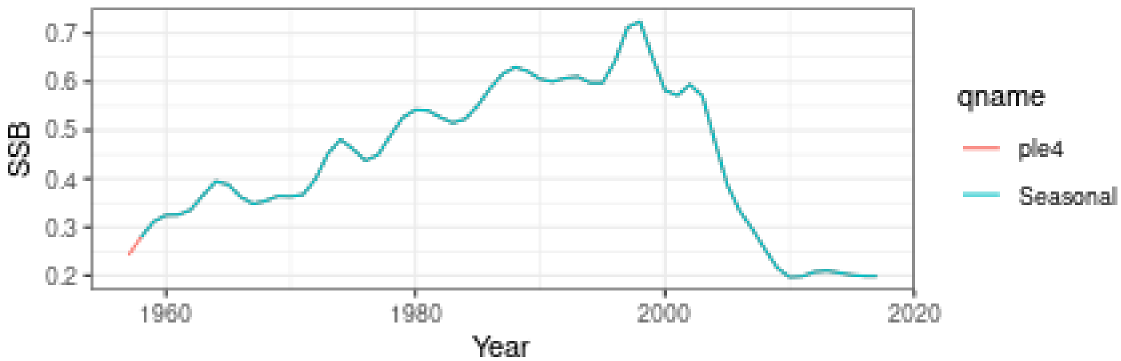
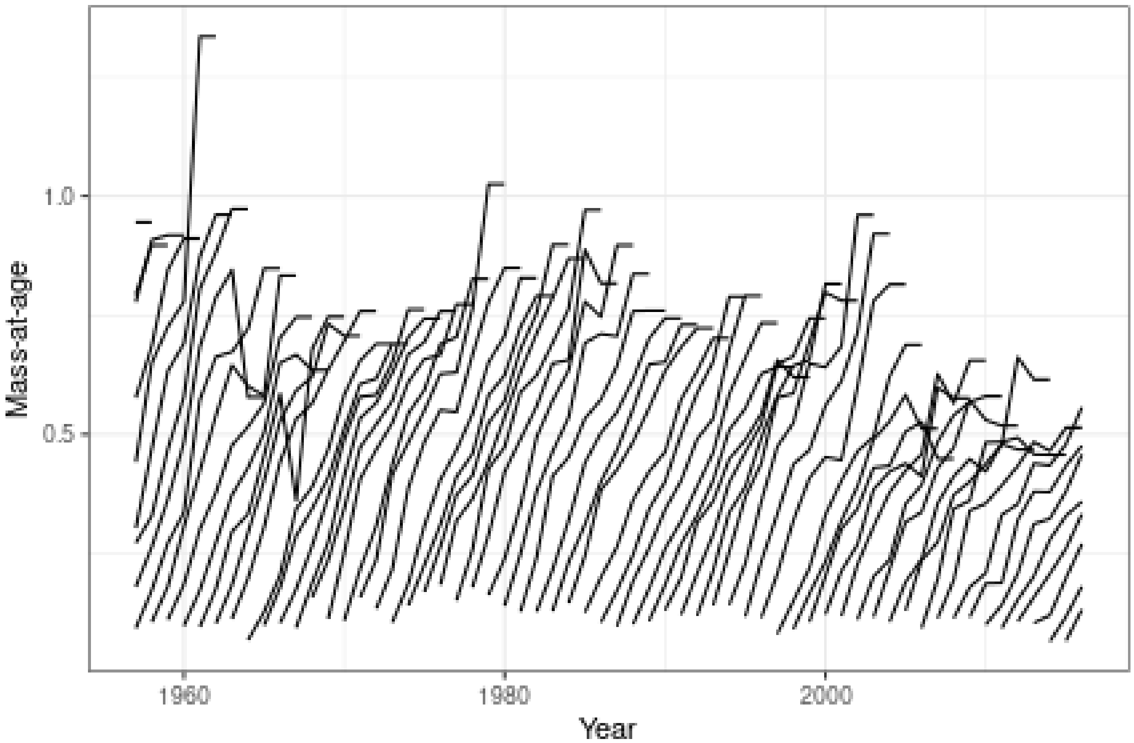


Figure 5 Comparison of Fs.

Seasonal growth can also be modelled

```
wtInterp<-function(wt){

  mlt=wt[,-dim(wt)[2]]
  mlt=FLQuant(rep(seq(0,(dim(mlt)[4]-1)/dim(mlt)[4],1/dim(mlt)[4]),each=max(cumprod(dim(mlt)[1:3]))),
             dimnames=dimnames(mlt))[-dim(mlt)[1]]

  incmt=wt[,,,1]
  incmt=-incmt[-dim(incmt)[1],-dim(incmt)[2]]+incmt[-1,-1]

  wt[dimnames(incmt)$age,dimnames(incmt)$year]=
    wt[dimnames(incmt)$age,dimnames(incmt)$year]+
    incmt%+%(mlt%*%incmt)

  wt[,-dim(wt)[2]]}
```



Projections

Now that the season model is consistent with the annual model it can be used to evaluate advice.

Extend years into the future

```
ple4.4=fwdWindow(ple4.4,end=2050,nsq=3)
```

To model and recruitment, requires specifying that the stock-recruitment object has dims for all seasons, but only values in the spawning season

```
sr        =predictModel(model=model(ple4.sr), params=params(ple4.sr))
params(sr)=FLPar(rep(c(params(sr)),4), dimnames=list(params=c('a','b'),season=seq(4), iter=1))
params(sr)[,-1]=NA
```

In non-spawning seasons maturity can be set to ‘NA’ or 0, although recruitment is determined by the ‘FLSR’ object.

Recruitmemt decviates can also be modelled, i.e. by providing recruitment deviates based on the annual model for the spawning season, others are ignore and can be set to ‘NA’ if desired.

Then set F_{bar} equal to $0.25F_{MSY}$ in each season

```
control=fwdControl(
  lapply(2017:2050, function(x) list(year=x, quant="fbar", value=computeRefpts(ple4.eq)["msy","harvest"]
```

To derive the expected values project with recruitment deviates set to 1, so there is no stochastic variability

```
mat(ple4.4)[is.na(mat(ple4.4))]=0

ple4.4.msy=fwd(ple4.4, control=control, sr=sr)
```

The F in the projection is equal to F_{MSY}, and the stock reaches B_{MSY} after about 20 years, while landings are at the MSY level.

```
plot(ple4.4.msy, metrics=list(SSB=function(x) ssb(x)[,,,1], Catch=landings, F=fbar)) +
   geom_flpar(data=FLPars(SSB   =FLPar(BMSY=refpts(ple4.eq)["msy","ssb"]),
                          Catch=FLPar( MSY=refpts(ple4.eq)["msy","yield"]/4),
                          F     =FLPar(FMSY=refpts(ple4.eq)["msy","harvest"]/4)), x=as.POSIXct("2011-01-0
  theme_bw()+
  xlab("Year")
```

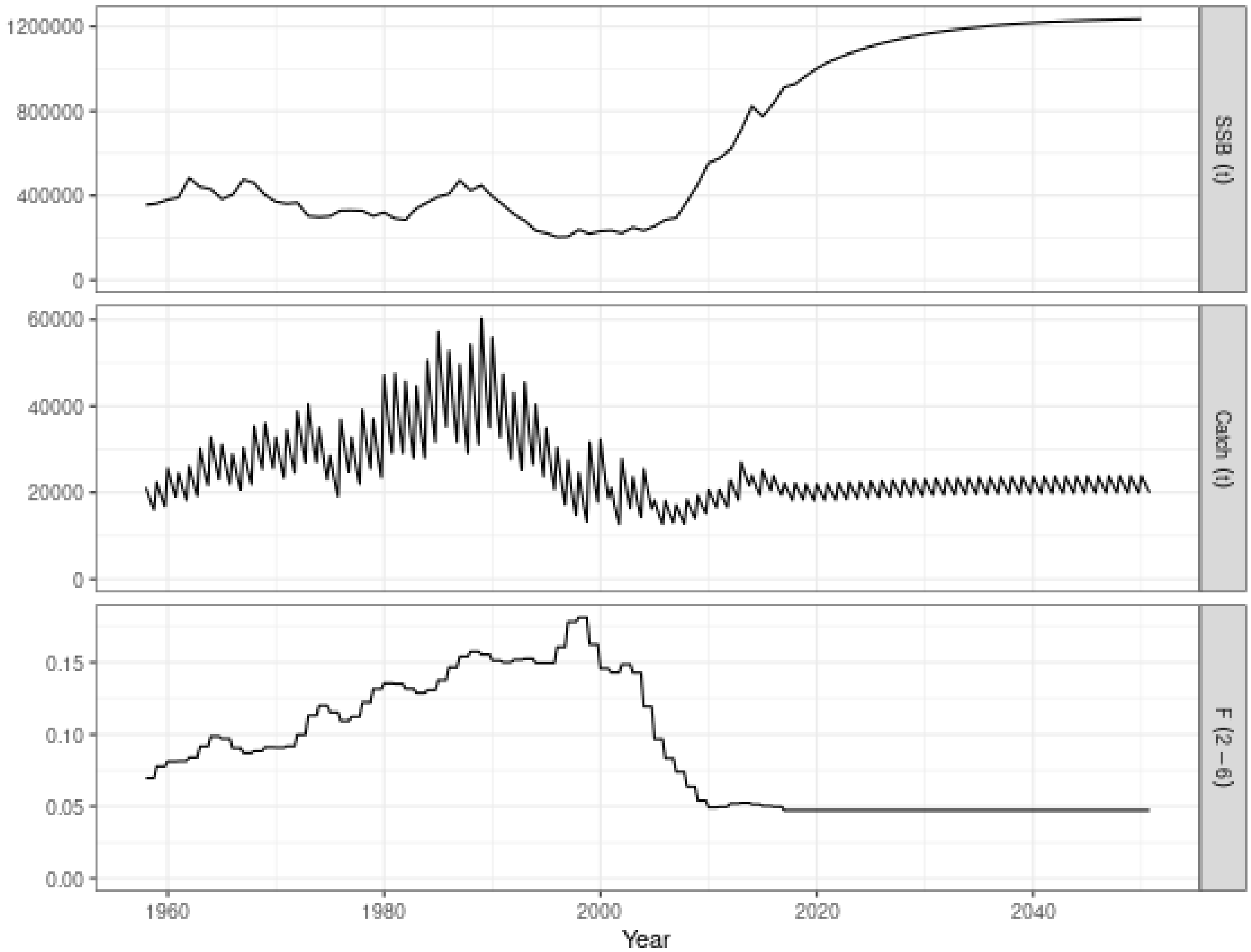


Figure 6 Projection.

Pelagic stock that recruits at age 0 and spawns once in season 2

The ‘seasonalise’ function automates adding seasons

```
load("neamac.RData")
```

```
plot(FLStocks("Annual"=neamac,"Seasonal"=seasonalise(neamac)[,-1]))
```

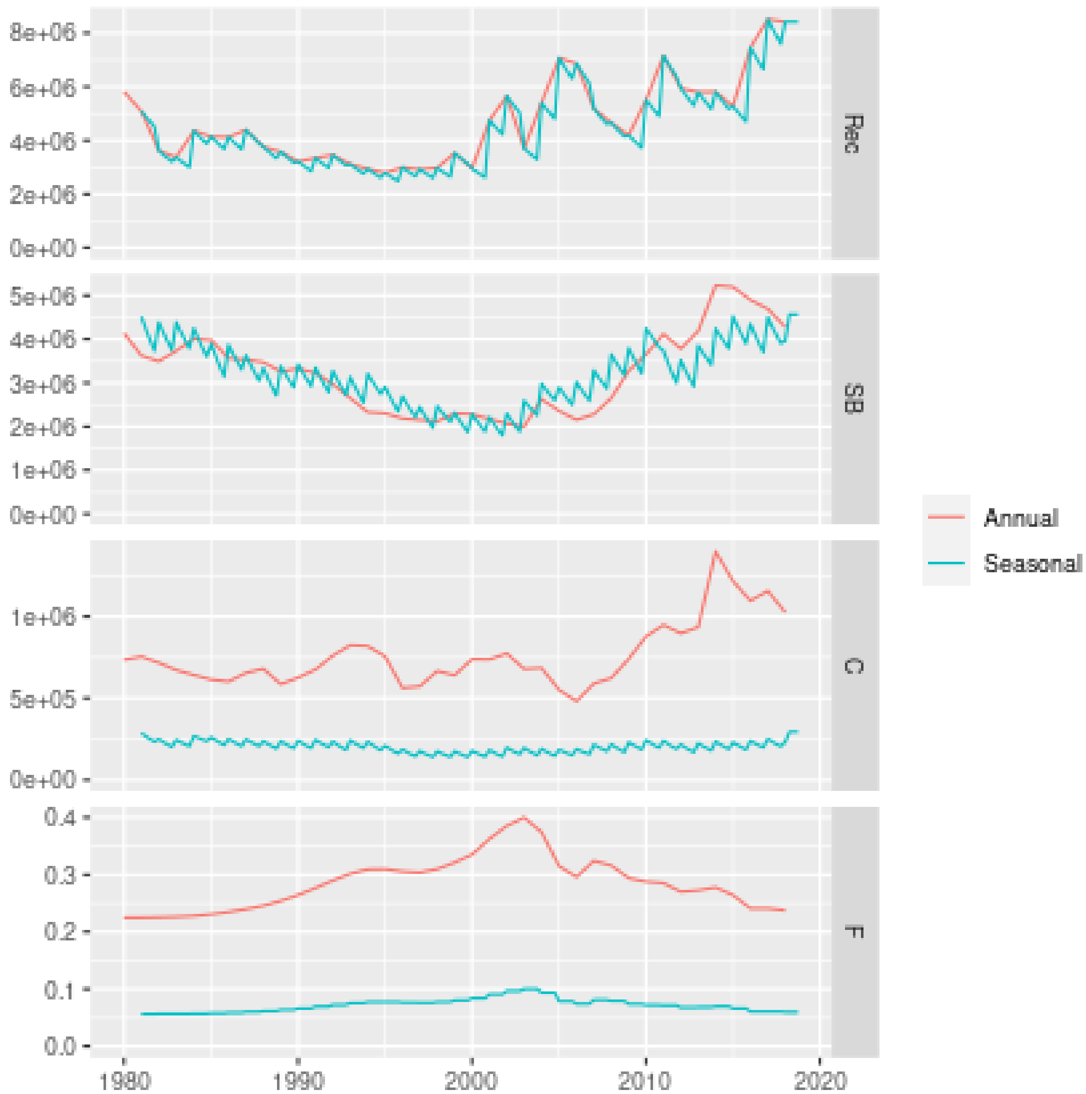


Figure 7 Comparison of annual and seasonal model

Albacore

The albacore stock already has seasons

```
load("alb.RData")
```

Set up projection in the form of a hindcast

```
mat(alb)[is.na(mat(alb))]=0

sr=as.FLSR(alb,model="geomean")
params(sr)=FLPar(1,dimnames=list(params="a",iter=1))
rec.devs=rec(alb)

control <- as(FLQuants(fbar=unitMeans(fbar(alb)[,-1])*c(rep(1,50),rep(0.5,dim(fbar(alb))[2]-51))), 'fwd(

alb.fwd <- fwd(alb, control=control, sr=sr, residuals=rec.devs)
```

```
plot(FLStocks("Original"=alb, "Recent F drop"=alb.fwd))
```

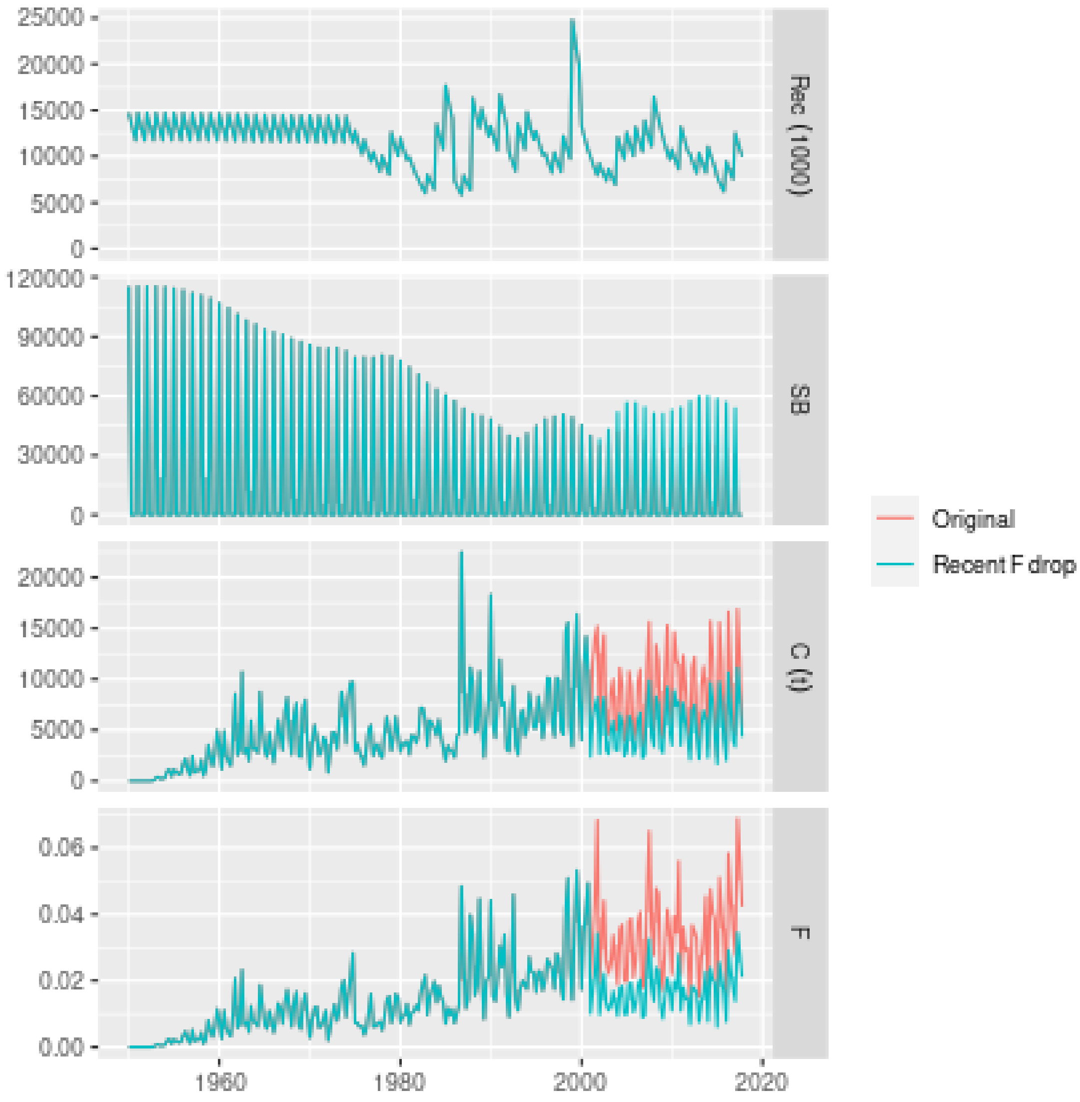


Figure 8 Backtest for reduced F

Tropical tuna, which spawns every season

Get SS3 object and perform hindcast to initialise

```
library(ss3om)

ss.file ="/home/laurence-kell/Desktop/projects/mrag/climate/task1/data/iccat/bet/2018/1-steepness0.7_MR
ss      =SS_output(ss.file,covar=FALSE)
rec.dist=ss$recruitment_dist$recruit_dist$Value
bet     =readFLSss3(ss.file)
bet.sr  =readFLSRss3(ss.file)

harvest(bet)[is.nan(harvest(bet))] <- 0
```

Recruitment is by year, unit & season, just provide the estimated recruitments

```
load("~/Desktop/flr/FLCandy/examples/seasonal/bet.RData")

sr=FLPar(NA, dimnames=list(params='a', year=1951:2017, unit=1:4, season=1:4, iter=1))

# Set spawning seasons
sr['a', , 1, 1]  <- c(stock.n(bet)[1, -1, 1, 1, drop=TRUE] )
sr['a', , 2, 2]  <- c(stock.n(bet)[1, -1, 2, 2, drop=TRUE] )
sr['a', , 3, 3]  <- c(stock.n(bet)[1, -1, 3, 3, drop=TRUE] )
sr['a', , 4, 4]  <- c(stock.n(bet)[1, -1, 4, 4, drop=TRUE] )

rec=predictModel(model=rec~a, params=sr)

# project for obsevered F
control=as(FLQuants(fbar=unitMeans(fbar(bet)[,-1])), 'fwdControl')

bet.fwd=fwd(bet, control=control, sr=rec)

p=plot(bet.fwd, metrics=list(rec  =function(x) rec(x),
                             SSB  =function(x) ssb(x),
                             fbar =function(x) apply(fbar(x), c(2,3),mean),
                             catch=function(x) apply(catch(x),c(2,3),sum)))
p$data=subset(p$data, !(qname%in%c("rec","SSB")&ac(season)!=ac(unit)))
p
```

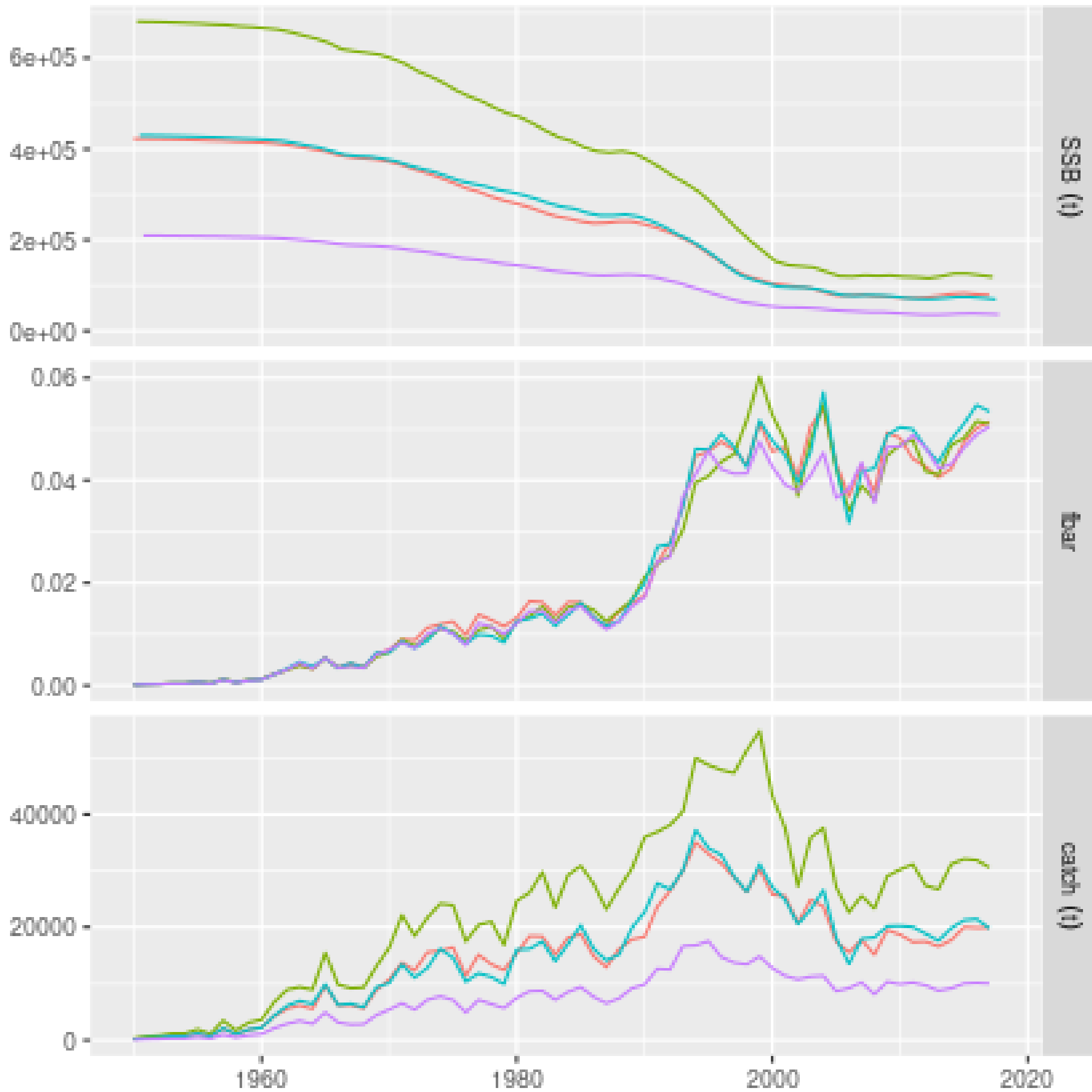


Figure 9 Atlantic bigeye tuna

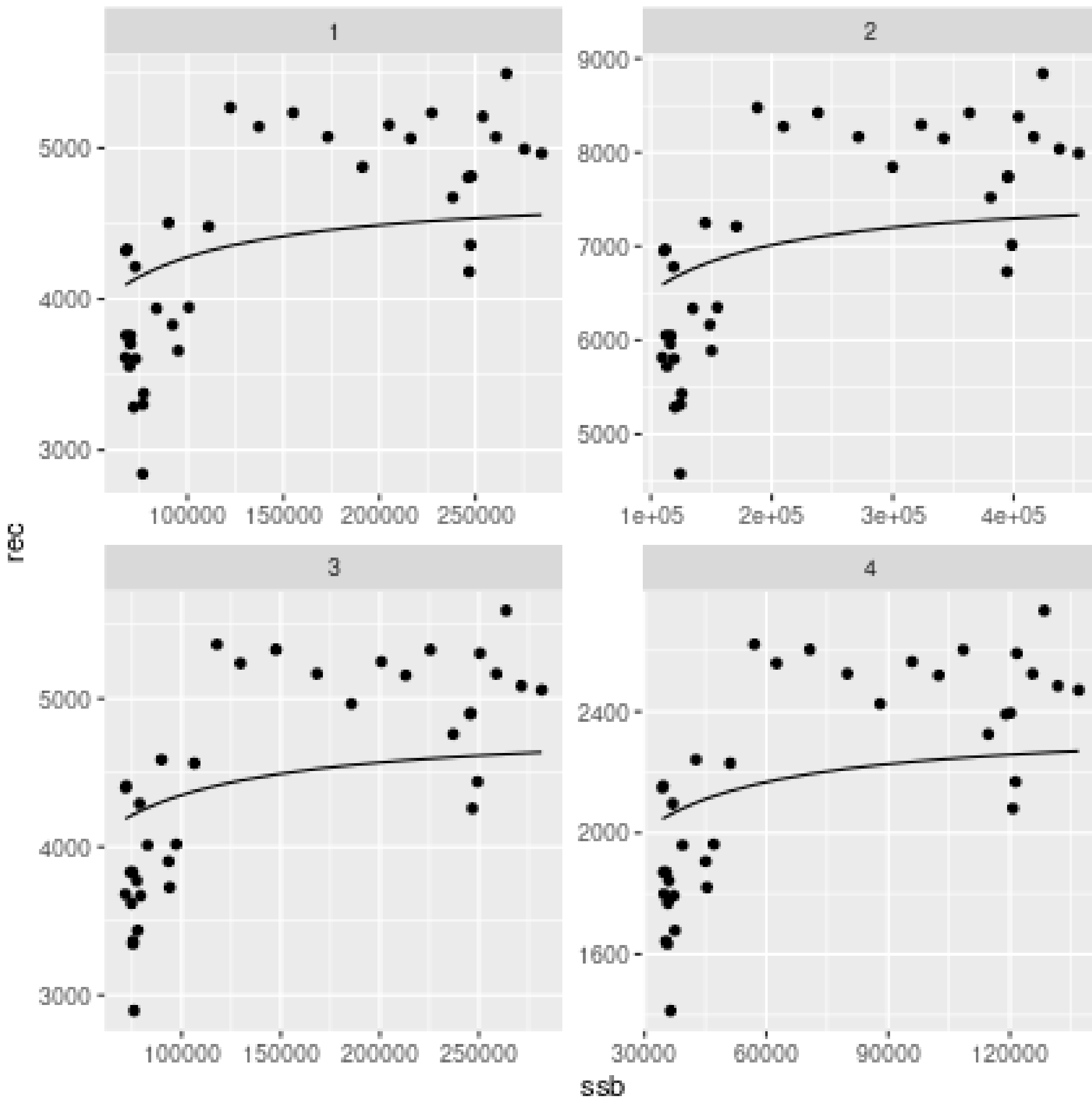
Perform a projection for a TAC for 65K tons

```
library(plyr)
library(dplyr)

params(bet.sr)["v"]/params(bet.sr)["R0"]
```

```
An object of class "FLPar"
params
   v
79.6
units:  NA
```

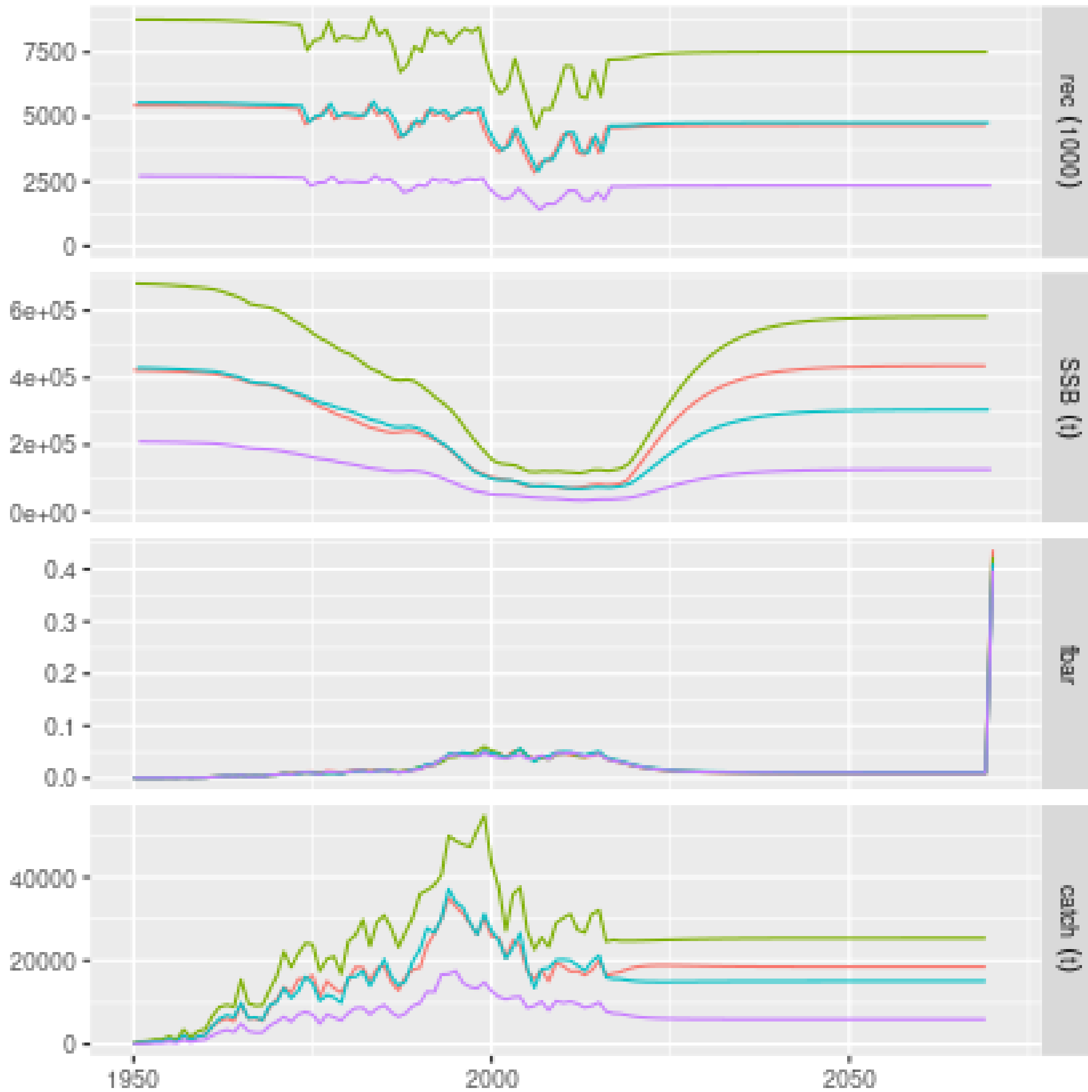
```
## Time series of recruits and SSB by season/unit
bet=window(bet,start=1980)
dat=merge(transmute(subset(as.data.frame(stock.n(bet)[1],drop=T),
                           season==unit),year=year,unit=unit,rec=data),
          transmute(subset(as.data.frame(ssb(bet),drop=T),
                           season==unit),year=year,unit=unit,ssb=data),
```

```
      by=c("year","unit"))

## SRR
srs=dlply(subset(dat,year>1980),.(unit), with, {
  sr=FLSR(ssb=as.FLQuant(data.frame(year=year,data=ssb)),
          rec=as.FLQuant(data.frame(year=year,data=rec)),model="bevholtSV")
  sr=fmle(sr,fixed=list(spr0=79.6,s=0.9),control=list(silent=TRUE))})

dat=merge(dat,ldply(srs, function(x) as.data.frame(predict(x),drop=T)))

ggplot(dat)+
  geom_point(aes(ssb,rec))+
  geom_line(aes(ssb,data))+
  facet_wrap(~unit,scal="free")
```




```
setMethod('rec', signature(object='FLStock'),
  function(object, rec.age=as.character(object@range["min"])){
    if(dims(object)$quant != 'age')
```

```
      stop("rec(FLStock) only defined for age-based objects")
    if(length(rec.age) > 1)
      stop("rec.age can only be of length 1")
    res <- stock.n(object)[rec.age,]
    return(res)
  })

sr=laply(srs, function(x) params(ab(x))[-3])
sr=FLPar(sr, dimnames=list(params=c('a','b'), season=1:4, iter=1))
sr=predictModel(model=bevholt()$model, params=sr)

ratio=apply(catch(bet)[,ac(2013:2017)],4,mean)/sum(apply(catch(bet)[,ac(2013:2017)],4,mean))
ratio=FLCore:::expand(ratio,year=1:54,fill=T)
ratio[]=ratio[,1]

dimnames(ratio)$year=as.numeric(dimnames(ratio)$year)+2015

control=as(FLQuants("catch"=ratio*65000),"fwdControl")

bet.fwd=fwdWindow(bet.fwd, end=2070,  nsq=5)
bet.fwd=fwd(bet.fwd, control=control, sr =sr)

p=plot(bet.fwd, metrics=list(rec  =function(x) stock.n(x)[1],
                             SSB  =function(x) ssb(x),
                             fbar =function(x) apply(fbar(x),c(2,3),mean),
                             catch=function(x) apply(catch(x),c(2,3),sum)))
p$data=subset(p$data, !(qname%in%c("rec","SSB")&ac(season)!=ac(unit)))
p
```



7500
5000
2500
0
rec (1000)
6e+05
4e+05
2e+05
0e+00
SSB (t)
0.4
0.3
0.2
0.1
0.0
fbar
40000
20000
0
catch (t)
1950
2000
2050

Plaice sexual dimorphism

```
data("ple4sex")

ple4sex.4=seasonalise(ple4sex)

plot(ple4sex.4)
```

Figure 10 Plaice with sexes and seasons

```
ple4sex.4=expand(ple4sex.4,area=1:2)

stock.n(ple4sex.4)[,,1,2]=0
stock.n(ple4sex.4)[,,2,1]=0

catch.n(    ple4sex.4)[,,1,2]=0
catch.n(    ple4sex.4)[,,2,1]=0
landings.n(ple4sex.4)[,,1,2]=0
landings.n(ple4sex.4)[,,2,1]=0
```

```
discards.n(ple4sex.4)[,,1,2]=0
discards.n(ple4sex.4)[,,2,1]=0

sr=as.FLSR(object,model="geomean")
params(sr)=FLPar(1,dimnames=list(params="a",iter=1))

recs=rec(ple4sex.4)

#fbar  =as(FLQuants("fbar"=as.FLQuant(fbar(ple4sex.4)[,-1,-1])),"fwdControl")

control=fwdControl(
  lapply(dimnames(ple4sex)$year[-1], function(x) list(year=x, quant="fbar", value=0.1, season=1:4,area=

ple4sex.4=fwd(ple4sex.4,control=control,sr=sr,residuals=recs)
```

Getting rid of seasons

```
sim=deSeason(bet)

plot(sim)
```